

Introduction to Machine Learning

Machine learning is a branch of artificial intelligence in which systems learn patterns from data to make predictions or decisions.

Types of Machine Learning

Supervised learning uses labeled training data. Unsupervised learning finds patterns in unlabeled data. Reinforcement learning learns through interaction and feedback in the form of rewards.

Supervised Learning

Common supervised learning tasks are classification and regression. Classification predicts categories, while regression predicts continuous numerical values.

Training and Testing

A dataset is commonly divided into training and testing sets. The training set is used to learn model parameters, and the test set is used to evaluate performance on unseen data.

Features and Labels

Features are input variables used by a model. A label or target is the output value the model is expected to predict.

Overfitting and Underfitting

Overfitting occurs when a model learns training data too closely and performs poorly on new data. Underfitting occurs when a model is too simple to capture important patterns.

Model Evaluation

Classification models can be evaluated using accuracy, precision, recall, and F1 score. Regression models can be evaluated using measures such as mean squared error.